

Classification.

Support Vector Machine on Binary Classification

Binary Classification problem

Train Data: $(\tilde{x}_1, y_1), (\tilde{x}_2, y_2), \dots, (\tilde{x}_n, y_n)$ where $\tilde{x}_i \in \mathbb{R}^d$ and $y_i \in \{1, -1\}$.

Test Data: $\tilde{x}_1, \tilde{x}_2, \dots \in \mathbb{R}^d$.

Goal: predict label $1, -1$ for each x_i

linear separable problem



There exists an affine map hyper plane $\langle w, x \rangle - b$ separate data.

or.

$\exists w \in \mathbb{R}^d, b \in \mathbb{R}$, s.t. $\text{sign}(\langle w, x_i \rangle - b) = y_i, \forall i=1, 2, \dots, n$.

For linear separable problem, there could be multiple choice of w and b . A good choice for better generalization is to choose a plane that maximize the margin.

formulation of the problem as optimization prob.



• firstly for linear separable data, $\exists$ separating

plane $\langle w, x \rangle - b = 0$, $\begin{cases} \langle w, x_i \rangle - b \geq 0, & y_i = 1 \\ \langle w, x_i \rangle - b \leq 0, & y_i = -1 \end{cases}$

• $\begin{cases} \langle w, x_i \rangle - b = 1 \\ \langle w, x_i \rangle - b = -1 \end{cases}$ s.t. $\begin{cases} \langle w, x_i \rangle - b \geq 1, & y_i = 1 \\ \langle w, x_i \rangle - b \leq -1, & y_i = -1 \end{cases}$

• Norm distance between the two

$\Rightarrow$ they pass the boundary of each class

• Norm dist between is $\frac{2}{\|w\|_2}$.

$$d = \min_{x \in \text{plane}} \|y - x\|_2^2 = \min_{x \in \mathbb{R}^d, z \in \mathbb{R}} \|y - x\|_2^2 + \lambda (\langle w, x \rangle - b)^2 \Rightarrow d = \frac{d_{\min} |b - \langle w, w \rangle|}{\|w\|_2}$$

at line. $\langle w, x \rangle - b = 1$ and $\langle w, x \rangle - b = -1$, here $d = d_1 + d_2 = \frac{2}{\|w\|_2}$.

- we wish to maximize ~~this dist~~ this distance.

Thus we can formulate the classification objective as.

$$\max_{w \in \mathbb{R}^d, b \in \mathbb{R}} \frac{2}{\|w\|_2}, \text{ s.t. } \begin{cases} w^T x_i - b \geq 1 & \text{if } y_i = 1 \\ w^T x_i - b \leq -1 & \text{if } y_i = -1. \end{cases}$$

primal form of SVM.

$$\min_{w, b} \frac{1}{2} \|w\|_2^2, \text{ s.t. } y_i (w^T x_i + b) \geq 1, \quad i=1, \dots, n.$$

Remark: • the constrained condition is equivalent as before.

- This problem can be solved by constrained opt methods.

kernel SVM.

To ~~for~~ introduce non linearity in SVM, one has to use kernel trick. We first invest the duality of LOpt.

Dual - primal formulation.

Consider the following constrained opt.

$$\min f(x), \text{ s.t. } \begin{cases} f_i(x) \leq 0, & i=1, \dots, m \\ h_i(x) = 0 & i=1, \dots, p. \end{cases}$$

in the Lagrangian penalty methods, we construct Lagrangian $L: \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, D is the joint feasible set.

$$L = f(x) + \sum \lambda_i f_i(x) + \sum \nu_i h_i(x).$$

Def L dual function.

$g: \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ is called Lagrange dual function as

$$g(\lambda, \nu) := \min_{x \in D} L(x, \lambda, \nu).$$

when L is unbounded below, $g \rightarrow -\infty$.

Remarks: • the optimal value of the original problem p^* ,
upper bound. $\mathcal{L} \Rightarrow \mathcal{L}(x, \lambda, v) \leq p^*, \forall \lambda \geq 0, v$.

• The optimal value of the original problem p^*
can be expressed as. $p^* = \min_{x \in D} \max_{\lambda, v} \mathcal{L}$.

The second remark is valid through following argument.

$$\sup_{\lambda \geq 0, v} \mathcal{L}(x, \lambda, v) = \begin{cases} f_0(x), & x \in D \\ \infty, & \text{otherwise.} \end{cases}$$

as if $x \notin D$, then $\exists i$ s.t. $f_i(x) > 0$ or $\exists i$ s.t. $h_i(x) \neq 0$.

either case, can choose proper λ_i or v_i s.t. $\mathcal{L} \rightarrow \infty$

$$\text{thus } \min_x \max_{\lambda, v} \mathcal{L} = \min_{x \in D} f_0(x) = p^*.$$

Lagrange dual problem.

The min-max inequality

$f: W \times Z \rightarrow \mathbb{R}, W \subseteq \mathbb{R}^n, Z \subseteq \mathbb{R}^m$, then.

$$\min_{w \in W} \max_{z \in Z} f(w, z) \geq \max_{z \in Z} \min_{w \in W} f(w, z)$$

proof. Let $w^0 \in W$, then we have $\max_{z \in Z} f(w^0, z) \geq f(w^0, z'), \forall z' \in Z$.

next let $z^0 \in Z$, then $\min_{w \in W} f(w, z^0) \leq f(w', z^0), \forall w' \in W$.

$$\text{thus } \min_{w \in W} f(w, z^0) \leq f(w^0, z^0) \leq \max_{z \in Z} f(w^0, z).$$

let $h(z) = \min_{w \in W} f(w, z)$, and $g(w) = \max_{z \in Z} f(w, z)$.

then we have $g(w) \geq h(z), \forall w \in W, z \in Z$.

$$\text{thus } g(w) \geq \max_{z \in Z} h(z), \forall w \in W$$

$$\text{thus } \min_{w \in W} g(w) \geq \max_{z \in Z} h(z) \Rightarrow \min_{w \in W} \max_{z \in Z} f(w, z) \geq \max_{z \in Z} \min_{w \in W} f(w, z).$$

Applying this inequality to the primal problem,
we got the Lagrangian Dual problem.

Def 1 Dual problem.

Let L be the Lagrangian of the primal problem, the corresponding dual problem is.

$$\max_{\lambda \geq 0, v} g(\lambda, v).$$

Remarks: from Th, we know the primal-dual is related with

$$p^* = \min_{x \in \mathcal{X}} \max_{\lambda \geq 0, v} L(x, \lambda, v) \geq \max_{\lambda \geq 0, v} \min_x L(x, \lambda, v) = d^*$$

meaning the optimal of the dual problem, is the upper bound of primal problem.

weak duality & duality gap.

when $d^* < p^*$, we called weak duality.

$p^* - d^*$ called duality gap.

Strong duality & Slater condition.

when $d^* = p^*$ holds, it called strong duality. specific.

condition must holds for strong duality, called constraint qualification. One simple example is Slater condition

~~Prop~~ - Slater condition

For primal problem $\min_x f_0(x)$, s.t. $f_i(x) \leq 0, i=1, \dots, m$
 $Ax = b.$

The Slater condition holds if

- $f_0, f_1, \dots, f_m$ are convex
- $\exists x \in D$, s.t. $f_i(x) < 0, i=1, \dots, m, Ax = b.$

back to SVM objective, it is not hard to see that the primal SVM satisfy Slater condition. Thus we can find the Dual form of SVM using the Lagrangian.

$$L = \frac{1}{2} \|w\|_2^2 + \sum_i \lambda_i (-(\langle w, x_i \rangle - b) y_i + 1)$$

we first find the $\min_{w \in \mathbb{R}^d, \lambda_i \in \mathbb{R}} L$ for Lagrangian dual function.

$$\begin{cases} \frac{\partial L}{\partial w_i} = w_i - \sum_j \lambda_j x_j^i y_j \\ \frac{\partial L}{\partial b} = \sum_i y_i \lambda_i \end{cases} \Rightarrow \begin{cases} w^* = \frac{1}{2} \sum y_i \lambda_i x_i \\ \sum y_i \lambda_i = 0 \end{cases}$$

$$\min_{w, b} L = \begin{cases} \frac{1}{2} \|w^*\|_2^2 + \sum_i \lambda_i (-(\langle w^*, x_i \rangle - b) y_i + 1), & \sum y_i \lambda_i = 0 \\ -\infty, & \sum y_i \lambda_i \neq 0 \end{cases}$$

it's if $\sum y_i \lambda_i \neq 0$, then $\lim_{b \rightarrow \infty} -b \sum y_i \lambda_i = -\infty$.

$$\begin{aligned} \frac{1}{2} \|w^*\|_2^2 &= \frac{1}{2} \left(\sum_i \lambda_i y_i x_i \right)^T \left(\sum_j \lambda_j y_j x_j \right) \\ &= \frac{1}{2} \langle w^*, w^* \rangle = \frac{1}{4} \sum_{i,j} y_i y_j \lambda_i \lambda_j \langle x_i, x_j \rangle \\ \sum_i \lambda_i (-(\langle w^*, x_i \rangle - b) y_i + 1) &= -\sum_i \lambda_i \langle w^*, x_i \rangle y_i + \sum_i \lambda_i \\ &= -\frac{1}{2} \sum_{i,j} \lambda_i \lambda_j y_i y_j \langle x_i, x_j \rangle + \sum_i \lambda_i \end{aligned}$$

$$\Rightarrow \min_{w, b} L = \begin{cases} -\frac{1}{4} \sum_{i,j} \lambda_i \lambda_j y_i y_j \langle x_i, x_j \rangle + \sum_i \lambda_i, & \sum y_i \lambda_i = 0 \\ -\infty, & \sum y_i \lambda_i \neq 0 \end{cases}$$

Dual form SVM

$$\max_{\lambda \geq 0} -\frac{1}{4} \sum_{i,j} \lambda_i \lambda_j y_i y_j \langle x_i, x_j \rangle + \sum_i \lambda_i$$

$$\text{s.t. } \sum y_i \lambda_i = 0$$

Remarks: In dual form, x_i appears as inner product, and can be replaced with kernel function.

Soft margin & non-separable set.

In many cases, the data are not linear separable (strictly). The solution is to introduce "soft" margin that allows misclassification. The constraints can be modified.

u.s.

$$y_i (\langle w, x_i \rangle - b) \geq 1 - \xi_i,$$

Remarks: instead of hard constraints that all class of data should be on one side of hyperplane.

Soft margins allows a set of hyperplanes that we would hope the ξ_i are not too large.

Primal problem with soft margin

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|_2^2 + C \sum \xi_i \\ \text{s.t.} \quad & y_i (\langle w, x_i \rangle - b) \geq 1 - \xi_i \\ & \xi_i \geq 0. \end{aligned}$$

$$\Leftrightarrow \begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|_2^2 + \frac{1}{C} \sum \xi_i \\ \text{s.t.} \quad & y_i (\langle w, x_i \rangle - b) \geq 1 - \xi_i \\ & \xi_i \geq 0. \end{aligned}$$

λ is regularization parameters

Dual problem with soft margin.

Following similar procedure of hard constraint problem, the dual form of soft margin is

$$\begin{aligned} \max_{\lambda \geq 0} \quad & \sum \lambda_i - \frac{1}{2} \sum_{i,j} \bar{y}_i y_j \lambda_i \lambda_j \\ \text{s.t.} \quad & \sum \lambda_i y_i = 0 \quad \& \quad \lambda_i \leq \frac{1}{2C\lambda}. \end{aligned}$$

Multi-class problem

The ~~decision~~ multi-class classification problem can be reduced to binary case. using '1 vs all' scheme.

For multi-class problem, we would have, data set as $(\bar{x}_1, y_1), \dots, (\bar{x}_n, y_n)$ as before, but $y_i \in \{0, 1, \dots, C-1\}$ for total of C distinct classes.

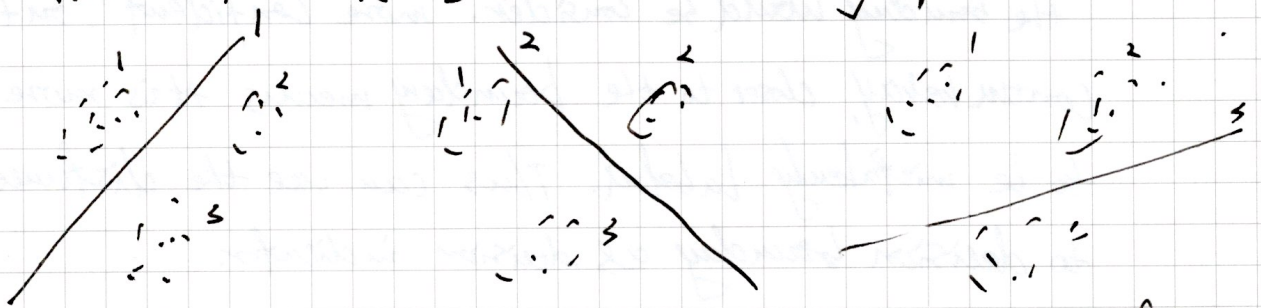
The goal of a multi-class classifier is ~~the~~ given a data point, it is in i th class, and not in all other classes. Therefore, ~~we can create~~ for each class, we can construct a binary classification problem: is the data point in this class or not. This will give a decision boundary between point in this class and those not. Do this for all classes, then we can obtain classification region based on these boundaries.

Example: 3-class $\Rightarrow$ we have three binary classifier for ~~first construct~~ class 1, 2, and 3.

For each binary classifier, we created the label as

$$\hat{y}_i = \begin{cases} 1, & \text{if } x_i \text{ in class } c \\ -1, & \text{if not} \end{cases} \quad c = 1, 2, 3.$$

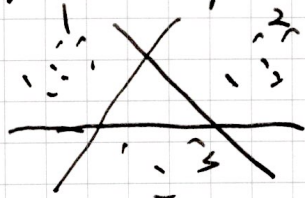
this will create 3 decision boundary for each class.



Each line separate the hyperspace into two region of $\in C$ and the other side $\notin C$. For before, if the weight for classifier of class c is $\vec{w}_c$, then we have.

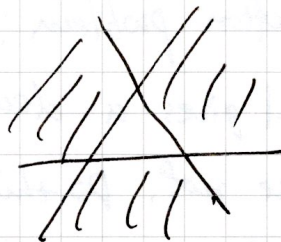
$$\begin{cases} x_i^T \vec{w}_c > 0 \rightarrow x_i \in C \\ x_i^T \vec{w}_c < 0 \rightarrow x_i \notin C \end{cases}$$

~~The first~~ this separate space into sub domains.



- ① The straightforward decision region is where the point only is on one positive side of a boundary, and on the negative side of all other boundary. then the predicted class is the class on the positive side. $\therefore$

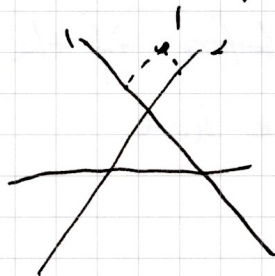
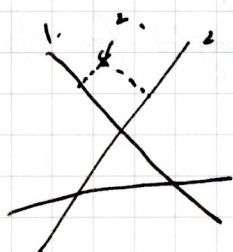
$$\hat{y}_i = \arg \max_c (x_i^T w_c - b_c)$$



- ② Region where ~~the~~ is the positive side of more than one boundary. means that the point can possibly belong to two class. Then we should choose the one that is more "plausible".

Note that for binary classifier, a point further from the boundary would be considered "more confident", on the contrary, closer to the boundary means it is more likely to be mistakenly labeled. Thus can use the distance to decision boundary as decision indicator.

Thus, the class for the point is the boundary that is furthest



the distance from x_i to boundary for class is

$$\frac{x_i^T w_c - b_c}{\|w_c\|_2}$$

$$\Rightarrow \hat{y}_i = \arg \max_{c=0, \dots, C} \frac{x_i^T w_c - b_c}{\|w_c\|_2}$$

③ region where all negative.



we can decide the class as the opposite of ② when a point is further from a boundary, means it is more confident not in that class, thus we

assign the class by the boundary with smallest distance.

note that $\frac{x_i^T w_c - b_c}{\|w_c\|_2}$ is the signed distance, thus

$$\hat{y}_i = \arg \min_c - \frac{x_i^T w_c - b_c}{\|w_c\|_2} = \arg \max_c \frac{x_i^T w_c - b_c}{\|w_c\|_2}$$

This 3-class example can be extended to general multi class case, and the procedure can be summarized as

input: $\{(\tilde{x}_i, y_i)\}$ for $y_i \in \{0, 1, \dots, C-1\}$.

for $i = 0, \dots, C-1$,

$$\tilde{y}_i = \begin{cases} +1 & \text{if } y_i = i \\ -1 & \text{else} \end{cases}$$

solve binary subproblem $\{(\tilde{x}_i, \tilde{y}_i)\} \rightarrow w_i, b_i$.

normalize weight $w_i, b_i \leftarrow \frac{w_i}{\|w_i\|_2}, \frac{b_i}{\|w_i\|_2}$.

get predicted label by $\hat{y} = \arg \max_c x^T w_c$.

Remark: • the normalization does not change for ①.

• in practice, can mark one-hot encoding

$$\begin{pmatrix} 1 \\ -1 \\ -1 \\ -1 \\ \vdots \end{pmatrix} \text{ if } i \in 1 \dots$$